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TECHNICAL SPECIFICATION: ENTERPRISE MULTI-LOCATION ARCHITECTURE & SECURE DATA STREAMING

1. Introduction & Strategic Purpose

This document specifies the enterprise-grade production architecture for the **SignageHub Modular Layout Engine**. Designed for large-scale corporate environments, high-end hospitality brands, and multi-location retail chains (e.g., enterprise coffee house operations spanning 100+ branches), this infrastructure ensures that localized content, regional parameters, and real-time operational database fields resolve dynamically at scale.

By treating individual screens as smart, location-aware nodes rather than hardcoded endpoints, SignageHub decouples content layout from physical geography. This specification details the mechanisms governing data streaming, network security compliance, network dropout recovery, and multi-tenant inheritance structures required for robust 24/7 continuous operation on unattended media player hardware.

2. Multi-Location Inheritance Architecture (Scaling Logic)

To eliminate the administrative overhead of configuring dozens or hundreds of displays individually, SignageHub enforces a strict **3-tier inheritance and placeholder model**. This structural pattern allows an enterprise network administrator to deploy a single global master playlist across an entire fleet while maintaining context-specific local data rendering.

Level 1: Hardware & Physical Screen Level (The Root Anchor)

Every active hardware player operating within the ecosystem registers a unique, immutable hardware identification token (Hardware ID) with the central cloud environment. Upon physical site deployment, the screen anchor is permanently mapped within the centralized database to its exact geographic coordinates using the **Google Maps API**. This registration locks the terminal's hardware instance to precise parameters: Latitude, Longitude, Regional Time Zone, and a localized Branch ID (e.g., `store_location_id: "lon_sub_042"`).

Level 2: Slide & Template Level (The Dynamic Placeholder Layer)

Content templates designed inside the SignageHub Layout Editor remain agnostic of physical geography. Instead of binding layouts to absolute locations or unique point-of-sale endpoints, layout modules accept native macro placeholders:

- Geographic Location Field: Resolved via [ LOCAL_SCREEN_LOCATION]
- Transactional Menu/Inventory Field: Resolved via [ LOCAL_STORE_INVENTORY]

This ensures that layout configurations remain reusable structural blueprints that inherit meaning only when executed on target hardware.

Level 3: Playlist & Distribution Level (The Fleet Deployment Engine)

The enterprise administrator generates a single master deployment playlist mapped to the unified 3x3 layout engine matrix. When this playlist is broadcast across the entire corporate fleet, individual media player terminals poll the backend independently. The cloud engine dynamically substitutes the macro placeholders with localized metrics corresponding to the calling unit's Branch ID before delivering the finalized configuration payload.

3. Secure POS & Menu Data Streaming (Backend-Mediated Flow)

To comply with strict enterprise IT security standards, including PCI-DSS guidelines, and to mitigate client-side browser performance penalties, frontend signage modules (such as `plugin.js` running on the media player) are completely isolated from third-party enterprise vendor infrastructure (e.g., Lightspeed, Toast, Trivec, Oracle MICROS Symphony). Direct frontend polling or client-side webhook endpoints on the display terminal are strictly prohibited.

The Secure Processing Lifecycle:

1. **Server-to-Server Authentication:** The **SignageHub App Server** establishes secure, backend-level OAuth handshake tunnels, encrypted webhook listeners, and authenticated REST connections with the client's point-of-sale (POS) cloud networks. No authentication tokens or secret keys reside on the public-facing hardware.
2. **Normalization & Aggregation:** Raw, highly nested inventory data structures and transactional events returned by third-party POS platforms are immediately parsed, sanitized, and stored inside SignageHub's relational data layer.
3. **Lightweight Frontend Polling:** The browser container on the physical skærm executes an asynchronous, lightweight fetch request targeting the native SignageHub endpoint on a safe 5-minute interval loop: `api.signagehub.tv/v1/screen-data?screen_id=XYZ`.
4. **Offline-Stable Resilience:** If the local area network encounters a dropout, a client-side Service Worker intercepts the request failure. Instead of crashing the presentation layer or prompting a browser error screen, the system reads from local IndexedDB storage (via `localStorage`), serving the last successfully cached data state to guarantee uninterrupted 24/7 continuous operation.

Standard Production JSON Payload Contract Structure:

The following data schema represents the sanitized, secure contract delivered to the target layout cell:

```
{
  "zone": "middle-center",
  "module": "MenuHUD",
  "theme": "hotel",
  "timestamp": 1779526500,
  "items": [
    { "name": "Espresso Blend Onyx", "status": "available", "meta": "Freshly Roasted" },
    { "name": "Single Origin Geisha", "status": "allocation", "meta": "ONLY 3 PACKAGES REMAINING" },
    { "name": "Cold Brew Nitro", "status": "sold-out", "meta": "Returns Tomorrow at 08:00" }
  ]
}
```

4. Design Integration & Symmetrical Rendering (Frontend Architecture)

Once the localized dataset resolves inside the terminal's sandboxed environment, native rendering engines enforce absolute visual alignment, preventing any spatial shifting that would compromise the high-end architectural look of the display.

- **The Unbreakable Internal Layout Grid:** To prevent highly variable strings or shifting item quantities from disrupting the surrounding 3x3 layout framework, all interactive data cards are governed by an internal grid enforcement layer. The core container injects inline styles preventing default block stacking: `display: grid !important; grid-template-columns: repeat(3, minmax(0, 1fr)) !important; gap: 15px !important;`. Cards adjust horizontally into perfect, non-overlapping 3-column matrices regardless of instance scale.
- **Intelligent "Sold Out" Masking:** When a specific inventory item shifts to an out-of-stock boolean state within the POS layer, the element is intentionally preserved within the DOM to maintain absolute spatial symmetry. A dedicated CSS modifier drops container opacity down to 20% and renders an elegant, thin cross-strikethrough text node: `[ SOLD OUT — RETURNING FOR TOMORROW'S MENU ]`. In Minimal Zen Mode, low-luminance typography preserves aesthetic cleanliness.
- **The Sommelier Allocation HUD:** When high-value inventory items cross below critical stock thresholds, the system automatically replaces price metadata with a high-priority "FOMO" string node: `[ ALLOCATION: ONLY X BOTTLES REMAINING ]`. In Luxury Hotel Mode, this transition triggers a hardware-accelerated CSS pseudo-element that introduces a slow, low-CPU ambient breathing glow mimicking soft candlelight.
- **Dynamic Edge-Anchoring (Transform Origins):** To ensure flawless UI scaling (0.5x to 1.5x) across varying professional commercial panels (from 1080p up to 8K), the layout manager overrides the elements' CSS `transform-origin` based on viewport zone coordinates. Top-tier sectors anchor strictly to the upper edge (`transform-origin: top;`), mid-tier to center, and bottom-tier to the lower screen boundary, forcing all content elements to clip cleanly against physical screen bezels during dimension adjustments.

Due Diligence & Compliance Note: This technical architecture blueprint complies with standard B2B enterprise isolation guidelines. By moving external API logic and authentication handshakes entirely to the server-side infrastructure, the client-facing presentation layer remains lightweight, secure, and immune to endpoint modifications executed by upstream platform providers.

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